VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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APROJECT REPORT

on

"CARDIOVASCULAR DISEASE PREDICTION USING RETINAL IMAGES"

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Engineering in computer Science and Engineering

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

The is to certify that the project entitled **"CARDIOVASCULAR DISEASE PREDICTION USING RETINAL IMAGES "** is a Bonafide work carried out by **RAJESHWARI B C (4CA22CS079)** in partial fulfillment for the degree of 7th Semester, Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya technological University, Belagavi during the year 2025-2026. It is certified that all corrections or suggestions indicated for Internal assessment have been incorporated on the report deposited in the department library. The major project report has been approved as it satisfies the academic requirements in respect of the major project prescribed for the Bachelor of Engineering Degree.

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ABSTRACT

Cardiovascular diseases (CVDs) remain a leading cause of global morbidity and mortality. Early detection and intervention are crucial for improving patient outcomes and reducing the burden on healthcare systems. Recent research suggests a potential link between retinal vascular changes and cardiovascular health. Retinal images offer a non-invasive means to assess microvascular abnormalities, making them an attractive source of data for predictive modeling.This project focuses on developing a machine learning model, specifically using Convolutional Neural Networks (CNNs), to analyze retinal images and detect patterns indicative of heart diseases. CNNs are particularly well-suited for image analysis as they can automatically learn spatial hierarchies of features, such as edges, textures, and shapes, which are critical for identifying abnormalities in retinal blood vessels. By leveraging CNNs, the project aims to enhance the accuracy and efficiency of heart disease detection, contributing to early diagnosis and timely intervention.

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CHAPTER 1

INTRODUCTION

Heart disease, also known as cardiovascular disease, refers to a class of diseases that involve the heart or blood vessels. It is a broad term encompassing various conditions that affect the cardiovascular system. The most common type of heart disease is coronary artery disease, which can lead to heart attacks.

This machine learning project aims to leverage Convolutional Neural Networks (CNNs) for the detection of heart diseases through the analysis of retinal images. The utilization of retinal images as a diagnostic tool has gained attention due to the potential correlation between retinal characteristics and cardiovascular health. The retina, as a neural tissue, shares vascular similarities with the cardiovascular system, and abnormalities in retinal vessels may indicate underlying heart conditions. The retina, located at the back of the eye, shares vascular similarities with the cardiovascular system. Abnormalities in the retinal blood vessels or microvascular changes may reflect systemic vascular health, including conditions related to the heart.

Convolutional Neural Networks (CNNs) are a class of deep learning models designed specifically for analyzing visual data. Unlike traditional neural networks, CNNs use convolutional layers to automatically extract spatial features such as edges, textures, and patterns from images, making them highly effective for image classification and recognition tasks. By employing CNNs, which excel in capturing spatial hierarchies within retinal images, this project seeks to improve the accuracy and efficiency of heart disease detection, contributing to early diagnosis and timely intervention.

This project holds significant importance in the healthcare domain as it provides a non-invasive and potentially cost-effective method for early detection of heart disease. If successful, the application of CNNs to retinal images could offer a proactive and scalable approach to cardiovascular health assessment, aiding healthcare professionals in timely decision-making and preventive care.

1.1 Objectives

The primary objectives of this machine learning project are to collect and preprocess a diverse dataset of retinal images annotated with cardiovascular health status, design and implement a Convolutional Neural Network (CNN)-based model for the analysis of retinal image data, optimize the model for sensitivity and specificity in heart disease detection, evaluate its performance using relevant metrics, and validate it on independent datasets. This approach aims to contribute to the early detection of heart diseases through retinal image analysis, ultimately improving patient outcomes and supporting informed clinical decision-making.

1.2 Problem Statement

Traditional methods of diagnosing heart diseases often rely on clinical parameters and expensive diagnostic tests. The lack of a non-invasive, cost-effective, and early screening tool poses challenges in timely disease detection. Retinal images, with their potential to reflect microvascular changes associated with cardiovascular conditions, present an opportunity to address this gap.

The challenge lies in developing a robust machine learning model that can effectively analyze retinal images and identify subtle patterns indicative of underlying heart diseases. Furthermore, addressing issues such as data variability, image quality, and interpretability is crucial for the model's clinical applicability.

1.3 Scope of the Project

1. Healthcare professionals could use the developed model as an auxiliary tool for early detection and screening of heart diseases during routine eye examinations.
2. The project could be integrated into telemedicine platforms, allowing for remote monitoring of patients' cardiovascular health through retinal images captured using portable devices or smartphones.

3. General practitioners and primary care physicians could benefit from the model as a supplementary diagnostic tool.
4. The project could be utilized in public health campaigns aimed at increasing awareness and promoting early detection of heart diseases.

1.4 Existing System

Traditional methods of detecting heart disease typically involve a combination of clinical assessment, medical history, physical examination, and various diagnostic tests. Healthcare professionals collect information about a patient's medical history, lifestyle, symptoms, and risk factors for heart disease. A thorough physical examination helps identify signs such as abnormal heart sounds (murmurs) or irregularities in the pulse. An ECG measures the electrical activity of the heart and can help identify irregularities in heart rhythm (arrhythmias), signs of a previous heart attack, and other cardiac abnormalities. Blood tests may be conducted to measure levels of cholesterol, triglycerides, and other lipids. Elevated levels of cholesterol and triglycerides can contribute to atherosclerosis, a major precursor to heart disease.

1.3.1 Disadvantages of Existing System:

1. Existing methods may lack the sensitivity to detect early stages of heart disease or subtle abnormalities, leading to delayed diagnosis.
2. Certain diagnostic tests, such as cardiac catheterization and angiography, involve invasive procedures that carry inherent risks and can cause discomfort to patients.
3. Interpretation of certain tests, such as stress testing or ECG, can be subjective and dependent on the skills and experience of the healthcare professional. This subjectivity may lead to variability in diagnoses.
4. Existing methods may lack the sensitivity to detect early stages of heart disease or subtle abnormalities, leading to delayed diagnosis.
5. Certain diagnostic tests, such as cardiac catheterization and angiography, involve invasive procedures that carry inherent risks and can cause discomfort to patients.

1.5 Proposed System

Convolutional Neural Networks (CNNs) are a type of deep learning model specifically designed to process and analyze visual data. Unlike traditional feedforward neural networks, CNNs employ convolutional layers that apply filters to input images, enabling the automatic extraction of spatial features such as edges, textures, and shapes. This hierarchical feature extraction makes CNNs particularly effective for tasks such as image classification, object detection, and medical image analysis.

Using CNNs with retinal image datasets involves leveraging their ability to capture local spatial relationships within images. By applying multiple convolution and pooling operations, CNNs can detect subtle variations in retinal blood vessels and other structures that may indicate cardiovascular abnormalities. This ability to automatically learn and extract meaningful features directly from retinal images eliminates the need for manual feature engineering, making CNNs a powerful tool for heart disease detection and other healthcare applications.

1.5.1 Advantages of proposed system are:

CNNs provide a robust framework for spatial understanding of retinal images. By capturing local patterns such as edges, textures, and shapes, CNNs can automatically identify subtle abnormalities in retinal blood vessels that may indicate cardiovascular risks. CNNs can handle retinal images of varying sizes and resolutions through techniques like convolution, pooling, and image resizing. This adaptability allows the model to work effectively with datasets collected from different imaging devices and sources. CNNs automate the feature extraction and pattern recognition process in retinal images, eliminating the need for manual feature engineering. This automation improves efficiency in large-scale screening and enhances diagnostic accuracy. CNNs are highly effective in detecting fine-grained spatial variations in retinal structures, which can correspond to early markers of cardiovascular disease. By learning hierarchical representations of image features, CNNs can provide precise and reliable predictions for heart disease detection.

CHAPTER 2

LITERATURE SURVEY

A literature review involves a thorough examination of existing published materials pertaining to a specific subject. This includes scholarly books, journals, and other relevant sources that relate to the chosen area of study. The central aim of conducting a literature review is to present to the reader the collective understanding and insights that have been generated regarding the chosen topic. This includes both the advantages and limitations associated with the existing knowledge.

1. Heart Attack Risk Prediction Using Retinal Eye Images Based On Machine Learning And Image Processing

Authors: Mrs. Sushma V, Sindhu K U, Sonupriya

Abstract: Heart disease increases the mortality rate in recent years across the world. So, it is necessary to develop a model to predict heart disease occurrence as early as possible with a higher rate of accuracy. Till now the detections are gone through blood tests, ECGs, and invasive stress tests. In this project, heart disease is predicted by a non- invasive method with the retinal image data. A Chase image dataset is considered, as the health of our eyes is connected to the health of our heart. Here, Heart problems can be detected from the changes in the microvasculature, which is imaged from the retina. The prediction of disease is by considering features like the size of blood vessels, non-uniform background illumination, etc. We use Image processing for identifying patterns in images and the Support Vector Machine (SVM) and Random Forest Classifier (RFC) algorithm for classification. The main objective of the proposed system is to predict the occurrence of heart disease from retinal fundus images with a higher rate of accuracy.

2. PREDICTION OF HEART DISEASE USING RETINAL IMAGES

Authors : Rupadevi,P. Sandhya Rani,K. Mounika,G. Lochana Bhavana,Ms. D. Ramya

Abstract: Heart disease increases the mortality rate in recent years across the world. So, it is necessary to develop a model to predict heart disease occurrence as early as possible with a higher rate of accuracy. Till now the detections are gone through blood tests, ECGs, and invasive stress tests. In this project, heart disease is predicted by a non- invasive method with the retinal image data. A Chase image dataset is considered, as the health of our eyes is connected to the health of our heart. Here, Heart problems can be detected from the changes in the microvasculature, which is imaged from the retina. The prediction of disease is by considering features like the size of blood vessels, non-uniform background illumination, etc. We use Image processing for identifying patterns in images and the Support Vector Machine (SVM) and Random Forest Classifier (RFC) algorithm for classification. The main objective of the proposed system is to predict the occurrence of heart disease from retinal fundus images with a higher rate of accuracy.

3. An Overview of Deep-Learning-Based Methods forCardiovascularRisk Assessment with Retinal Images

Authors: Rubén G. Barriada and David Masip.

Abstract: Cardiovascular diseases (CVDs) are one of the most prevalent causes of premature death. Early detection is crucial to prevent and address CVDs in a timely manner. Recent advances in oculomics show that retina fundus imaging (RFI) can carry relevant information for the early diagnosis of several systemic disease.There is a large corpus of RFI systematically acquired for diagnosing eye - related diseases that could be used for CVDs prevention. Nevertheless, public health systems cannot afford to dedicate expert physicians to only deal with this data, posing the need for automated diagnosis tools that can raise alarms for patients at risk.Artificial Intelligence (AI) and, particularly, deep learning models, became a strong alternative to provide computerized pre-diagnosis for patient risk retrieval.

This overview gathers commonly used datasets, pre- processing techniques, evaluation metrics and deep learning approaches used in 30 different studies. Based on the reviewed articles, this work proposes a classification taxonomy depending on the prediction target and summarizes future research challenges that have to be tackled to progress in this line.

4. Heart Disease Prediction Using Eye Retinal Image

Authors: Sameer Shaikh; Ishan Wagh; Vrushab Zaveri; Mohammad Bazil Mujawar;

Abstract: Cardiovascular disease (CVD) remains the leading cause of morbidity and mortality worldwide. Early detection and risk assessment of cardiovascular diseases are important for effective prevention and timely intervention. An analysis of deep learning methods used for prediction of CVR based on RFI. Since it is an extension of central nervous system, the retina gives an exceptional chance for safe medical diagnosis. Using multiple types of retinal fundus images together with the patient and clinical data, we developed a new deep learning model for prediction of several vital parameters for cardiovascular diseases. With CNN, it is possible to extract relevant features from retinal images without involving any manual procedures.

5. Predicting Cardiovascular Risk Factors from Retinal Fundus Photographs using Deep Learning

Authors: Ryan Poplin, MS Avinash V. Varadarajan Katy Blumer,Yun Liu,Michael V. McConnell

Abstract: Traditionally, medical discoveries are made by observing associations and then designing experiments to test these hypotheses. However, observing and quantifying associations in images can be difficult because of the wide variety of features, patterns, colours, values, shapes in real data. In this paper, we use deep learning, a machine learning technique that learns its own features, to discover new knowledge from retinal fundus images. Using models trained on data from 284,335 patients, and validated on two independent datasets of 12,026 and 999 patients, we predict cardiovascular risk factors not previously thought to be present or quantifiable

in retinal images, such as such as age (within 3.26 years), gender (0.97 AUC), smoking status (0.71 AUC), HbA1c (within 1.39%), systolic blood pressure (within 11.23mmHg) as well as major adverse cardiac events (0.70 AUC). We further show that our models used distinct aspects of the anatomy to generate each prediction, such as the optic disc or blood vessels, opening avenues of further research.

6. Use of artificial intelligence on retinal images to accurately predict the risk of cardiovascular event

Authors: Ehsan Vaghefi David Squirrell Song Yang Songyang An John Marshal

Abstract: Purpose: To create and evaluate the accuracy of an artificial intelligence platform capable of using only retinal fundus images to predict both an individual's overall 10 year Cardiovascular Disease (CVD) risk and the relative contribution of the component risk factors that comprise this risk (CVD-AI). Methods: The UK Biobank and the US-based AREDS 1 datasets were obtained and used for this study. The UK Biobank data was used for training, validation and testing, while the AREDS 1 dataset was used as an external testing dataset. Overall, we used 110,272 fundus images from 55,118 patient visits. A series of models were trained to predict the risk of CVD against available labels in the UK Biobank dataset. Results: In both the UK Biobank testing dataset and the external validation dataset (AREDS 1), the 10-year CV risk scores generated by CVD-AI were significantly higher for patients who had suffered an actual CVD event when compared to patients who did not experience a CVD event. In the UK Biobank dataset the median 10-year CVD risk for those individuals who experienced a CVD was higher than those who did not (4.9% [ICR 2.9-8%] v 2.3% [IQR 4.3- 1.3%] P<0.01.]. Similar results were observed in the AREDS 1 dataset The median 10-year CVD risk for those individuals who experienced a CVD event was higher than those who did not (6.2% [ICR 3.2%- 12.9%] v 2.2% [IQR 3.9-1.3%] P<0.01).

CHAPTER 3

SYSTEM SPECIFICATION

3.1 Hardware requirements

- Processor: Intel Core i3 or higher
- Storage: 10GB or higher
- RAM:8GB OR higher

3.2 Software requirements

- Operating System: Windows 10
- Frontend: HTML, CSS, Flask
- Backend: Python 3.10 or higher
- Database: MySQL

3.3 Functional requirements

The system has two main types of users: Admin and User. The Admin is primarily responsible for managing the training phase of the CNN model. Their tasks include uploading retinal image datasets, initiating the training process, evaluating performance, and saving the trained model for future use. On the other hand, the User interacts with the system during the testing phase. After registering and logging in, Users can upload their retinal images, which the system processes to predict the presence of heart disease risk. The simplicity of the User interface ensures that even individuals without technical expertise can utilize the system effectively. This clear division of roles between Admin and User ensures that the system remains organized, secure, and efficient.

- **Admin:** Admin has the responsibility of overseeing the training phase of the application. Once logged in successfully, Admin can upload the training dataset, initiate the training process using the ML algorithm, create a model, and view the training results.

Features

1. Login
2. Upload Dataset
3. Train
4. Save Model
5. View Performance

- **User:**The User has the responsibility of overseeing the testing phase of the application. He has to first register with his details. After successful login, he can input test image and check whether there is risk of heart attack or not.

1. Register
2. Login
3. Input Test Image
4. View Prediction

3.4 Non-Functional Requirements

Non-functional requirements for this project would encompass various aspects related to system performance, reliability, security, usability, and scalability. Here are some non-functional requirements that should be considered:

- **Performance:**The system should be capable of processing a large volume of retinal images efficiently to ensure timely diagnosis. The processing time per image should be within acceptable limits to maintain user satisfaction and minimize waiting times. The accuracy of the detection algorithm should meet or exceed industry standards.
- **Scalability:**The system should be designed to accommodate an increasing number of users and images without significant degradation in performance.
- **Reliability:**The system should exhibit high reliability to ensure consistent performance under normal operating conditions. It should be resilient to potential failures, such as hardware malfunctions or network outages, to minimize downtime and data loss.
- **Usability:**The user interface should be intuitive and easy to navigate.

3.5 Tools and Libraries Used

1.Python for Programming

Python is one of the most widely used programming languages in artificial intelligence and machine learning projects due to its simplicity, readability, and extensive support libraries. Its flexible syntax allows developers to focus on problem-solving rather than dealing with complex programming constructs. With a vast ecosystem of libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn, Python provides efficient tools for data preprocessing, visualization, and modeling. In this project, Python serves as the backbone for implementing the CNN algorithm, handling data operations, integrating with databases, and managing the overall application workflow.

2.Flask for Web Application Framework

Flask is a lightweight yet powerful Python-based web framework used for developing web applications with minimal overhead. It follows a modular design and adheres to the Model-View-Controller (MVC) pattern, which makes it flexible and easy to extend for custom requirements. In this project, Flask is responsible for connecting the trained CNN model with the user interface, handling HTTP requests, routing user inputs such as CT scan uploads, and returning prediction results. Its simplicity makes it an ideal choice for academic projects where quick prototyping, easy deployment, and smooth integration are essential.

3.TensorFlow/Keras and Resnet50 for CNN Model Development

TensorFlow, with its high-level API Keras, is one of the most widely used frameworks for deep learning and neural network implementation. It provides efficient tools for building, training, and evaluating deep learning models, including CNNs, which are particularly effective for image analysis tasks. In this project, TensorFlow/Keras is used to design and implement the CNN architecture, including convolutional, pooling, and dense layers. The framework’s flexibility allows experimentation with hyperparameters, optimization techniques, and visualization of training performance, ensuring the model achieves high accuracy and robustness.

ResNet-50 is a deep convolutional neural network architecture consisting of 50 layers and is widely used in medical image analysis due to its ability to learn complex features efficiently. In cardiovascular disease prediction using retinal images, ResNet-50 plays a crucial role in extracting microvascular patterns such as blood vessel thickness, tortuosity, hemorrhages, and lesions present in retinal fundus images. The key strength of ResNet-50 lies in its residual (skip) connections, which allow the network to bypass certain layers and prevent the vanishing gradient problem, enabling effective training of very deep networks. The architecture uses bottleneck residual blocks with 1×1, 3×3, and 1×1 convolutions to reduce computation while preserving important features. By leveraging transfer learning with a pretrained ResNet-50 model, the network can accurately capture subtle retinal changes that are strongly associated with cardiovascular conditions, making it a reliable and high-performance model for disease prediction.

4.MySQL for Database Management

MySQL is a popular relational database management system known for its reliability, scalability, and open-source nature. It is widely used in web-based applications for secure data storage and retrieval. In this project, MySQL is used to manage user authentication, store registration details, and log administrative tasks such as dataset uploads and training results. Its integration with Python via connectors ensures seamless interaction between the web application and the database. By handling structured data efficiently, MySQL supports the backend operations, ensuring smooth and secure workflow management.

5.Python IDLE for Development

Python IDLE (Integrated Development and Learning Environment) is a simple, beginner-friendly IDE that comes bundled with Python. It provides a clean interface for writing, running, and debugging Python scripts, making it especially useful in academic projects. In this project, Python IDLE is used for developing and testing smaller code modules such as preprocessing functions, CNN model training routines, and database connectivity scripts before integrating them into the larger system. Its built-in interactive shell allows quick prototyping and testing, which accelerates the development process.

CHAPTER 4

SYSTEM DESIGN

4.1 System Perspective

The project is designed as a web-based application built using Python and Flask, integrated with TensorFlow/Keras for CNN model development and MySQL for database management. From the system's perspective, it functions as a two-part framework: the Admin side and the User side. The Admin side focuses on data collection, preprocessing, and model training, while the User side emphasizes image input and prediction generation. The application ensures seamless data flow between modules—ranging from image upload, preprocessing, model execution, and prediction display—supported by a secure database. This modular design provides scalability, reliability, and maintainability, making the system adaptable to real-world healthcare needs.

4.1.1 System Definition

The system is defined as a CNN-powered web application that detects the risk of heart disease using retinal images. It consists of two functional modules: Admin Module and User Module. The Admin Module enables dataset management, training, and evaluation of the CNN model, ensuring that the system stays updated with reliable predictions. The User Module allows individuals to register, log in, upload retinal images, and receive immediate feedback regarding heart disease risk. The system is further supported by a backend database for secure storage of user information and training records. Overall, the system definition emphasizes a non-invasive, cost-effective, and efficient diagnostic support tool for cardiovascular health monitoring.The Admin Module enables dataset management, training, and evaluation of the CNN model, ensuring that the system stays updated with reliable predictions. The User Module allows individuals to register, log in, upload retinal images, and receive immediate feedback regarding heart disease risk.

4.2 System Architecture

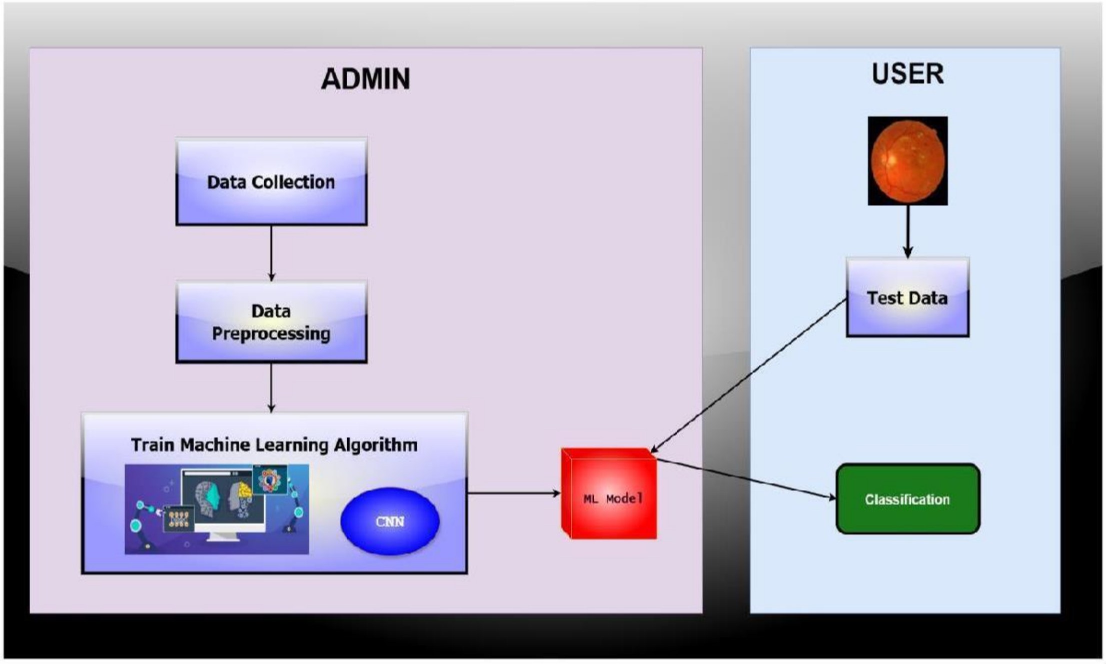

Fig 4.1 System architecture

A system architecture is the conceptual model that defines the structure, behavior, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system.

This diagram shows a system for detecting heart disease using retinal images, divided into two parts: ADMIN and USER. The ADMIN side involves collecting and preprocessing data, then training a machine learning model (indicated as CNN). Once trained, this model is ready to analyze new images. On the USER side, user upload retinal images, which the trained model then classifies to determine the presence of heart disease, providing a diagnostic result based on the analysis.

4.3 UML Diagrams

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented

Computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modelling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing object-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

4.3.1 Use Case Diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor.

Use case – Admin

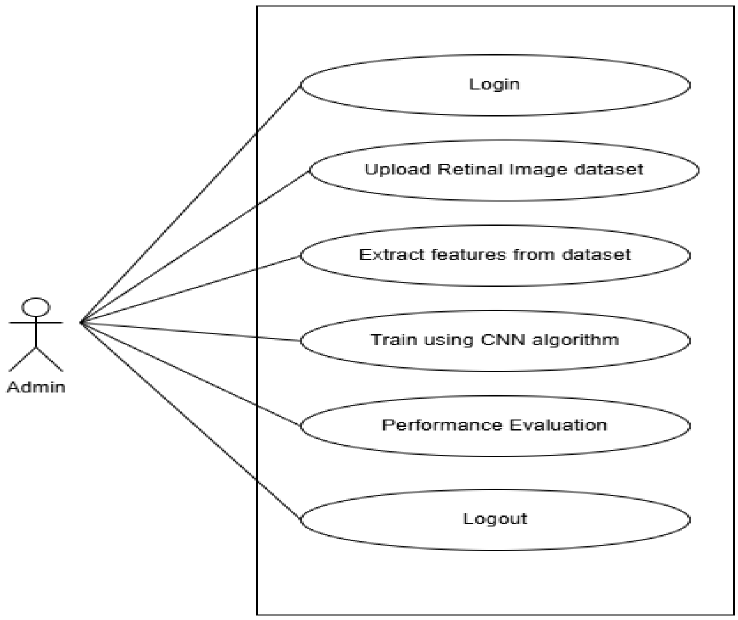

Fig 4.2 Use case of admin

This use case diagram illustrates the interactions an admin has with a heart disease detection system using retinal images. The admin logs in, uploads a retinal image dataset, extracts features from the dataset, and then trains a machine learning model (specifically using CNN algorithm). After training, the admin evaluates the model's performance to ensure it is accurate and reliable. Once these tasks are complete, the admin logs out. This diagram captures the essential steps for managing and improving the machine learning model within the system.

Use case – User

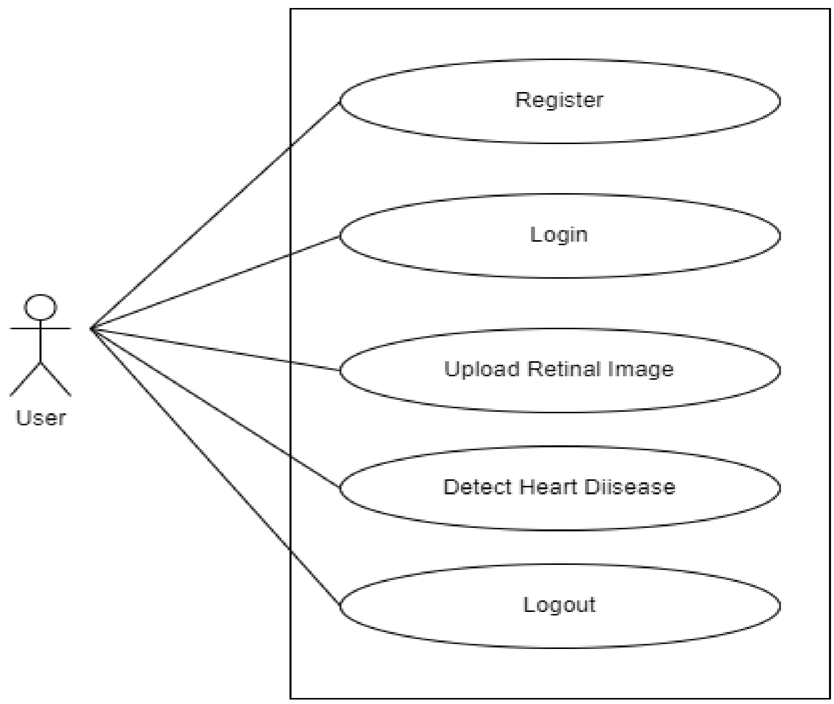

Fig 4.3 Usecase of user

4.3.2 Sequence Diagram:

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

The diagram illustrates the interactions between the user and the system. The user, represented by a stick figure, can perform several actions (use cases) within the system, which are depicted as ovals. The available actions include registering for an account, logging into the system, uploading a retinal image, detecting heart disease.from the uploaded retinal image, and logging out. Each use case is connected to the user, indicating that the user can initiate these actions.

Sequence – Admin

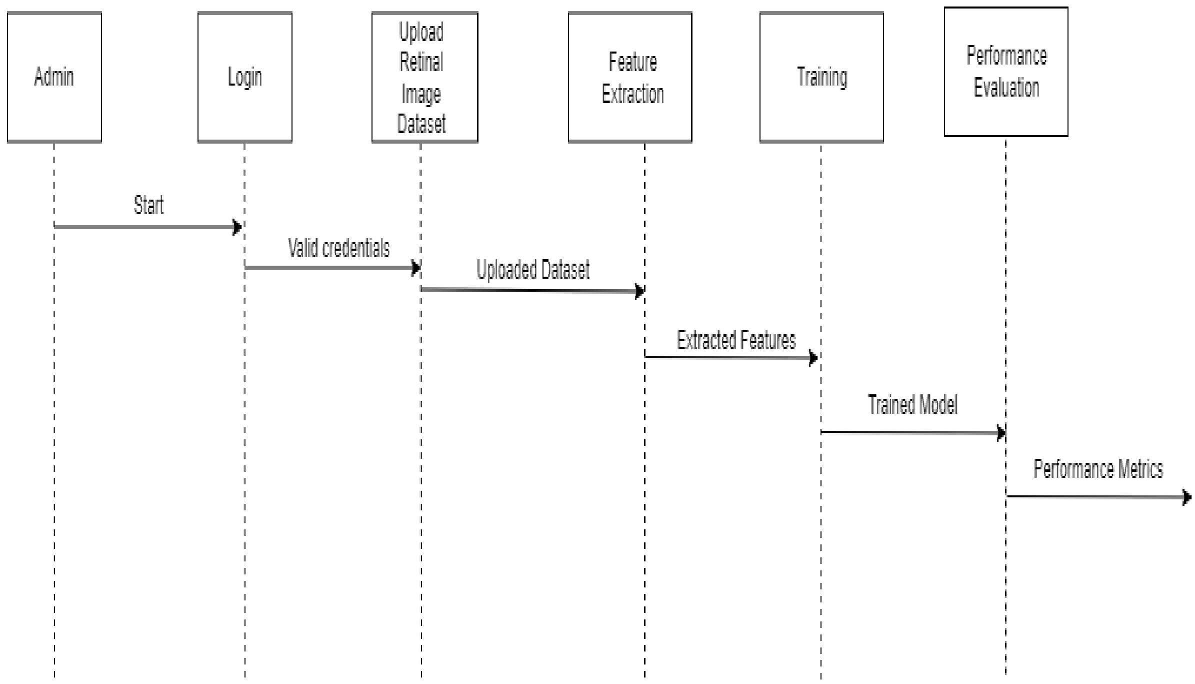

Fig 4.4 Sequence diagram of admin

The diagram is a sequence diagram for the administrative workflow of a heart disease detection system. It outlines the steps involved from logging in to evaluating the system's performance. The process begins with the admin logging into the system, provided valid credentials. Upon successful login, the admin uploads a retinal image dataset. This dataset undergoes feature extraction, where important characteristics are identified. The extracted features are then used to train a machine learning model. Finally, the performance of the trained model is evaluated, resulting in performance metrics that indicate the effectiveness of the model in detecting heart disease. The sequence of actions and data flow between each step is shown by the horizontal arrows, indicating the progression from one phase to the next.

Sequence – User

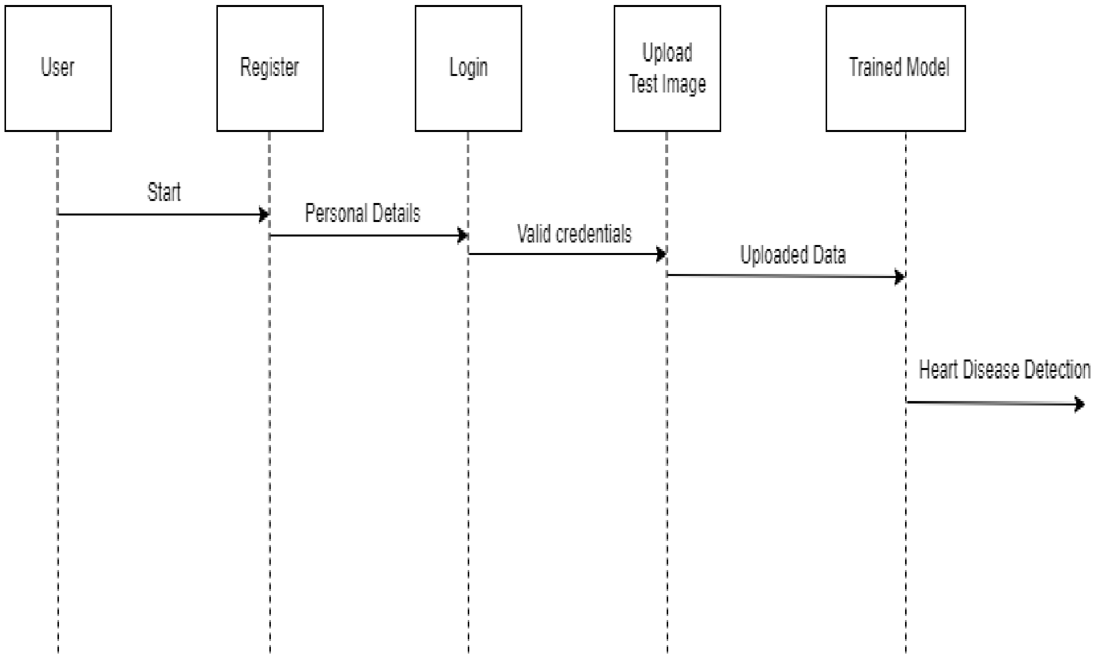

Fig 4.5 Sequence diagram of user

The diagram is a sequence diagram for the administrative workflow of a heart disease detection system. It outlines the steps involved from logging in to evaluating the system's performance. The process begins with the admin logging into the system, provided valid credentials. Upon successful login, the admin uploads a retinal image dataset. This dataset undergoes feature extraction, where important characteristics are identified. The extracted features are then used to train a machi ne learning model. Finally, the performance of the trained model is evaluated, resulting in performance metrics that indicate the effectiveness of the model in detecting heart disease. The sequence of actions and data flow between each step is shown by the horizontal arrows, indicating the progression from one phase to the next.

4.3.3 Activity Diagram:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency.

In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

Activity Diagram – Admin

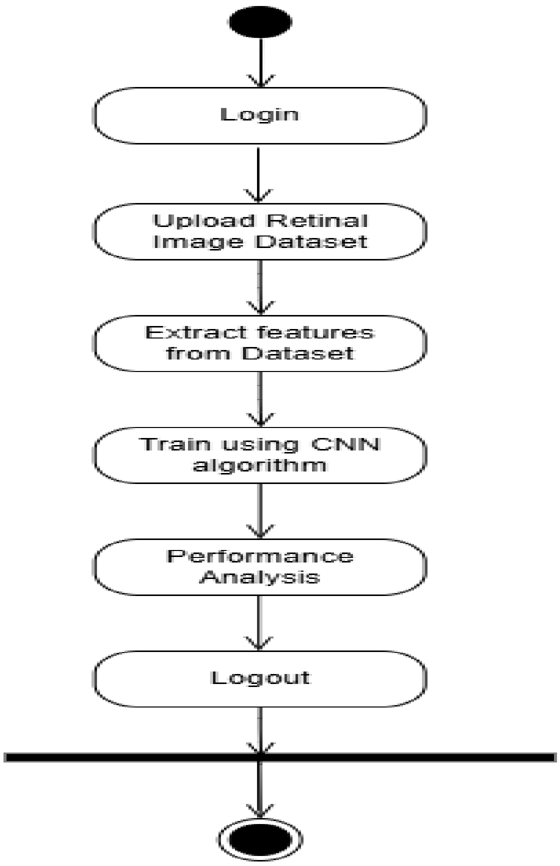

Fig 4.6 Activity diagram of admin

This activity diagram represents the workflow of a heart disease detection system using retinal images. The process begins with the user logging into the system. After logging in, the user uploads a retinal image dataset. The next step involves extracting features from the dataset, which are then used to train a model using a Convolutional Neural Network (CNN) algorithm. Following the training, a performance analysis is conducted to evaluate the model. Finally, the user logs out of the system, completing the activity sequence.

Activity Diagram – User

This activity diagram outlines the workflow for a user interacting with a heart disease detection system based on retinal images. The process starts with the user registering on the system. After registration, the user logs in and proceeds to upload a retinal image. The system then processes this image to detect the presence of heart disease. Once the detection process is complete, the user can log out, thus finishing the interaction with the system.

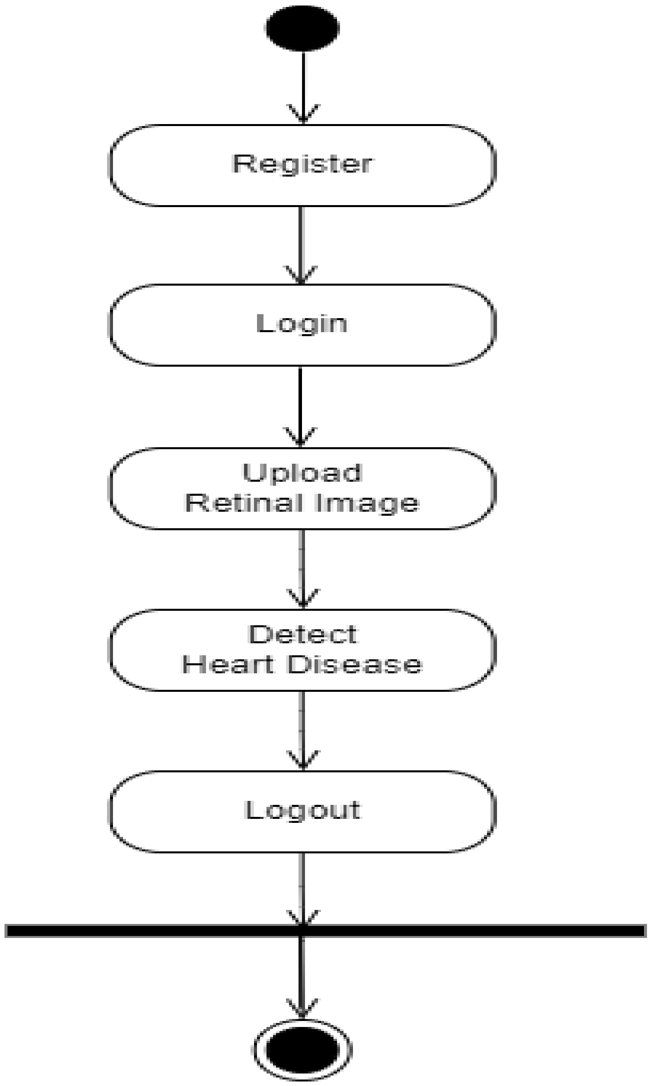

Fig 4.7 Activity diagram of user

CHAPTER 5

IMPLEMENTATION

The proposed system will be developed using Python and Flask, a web application framework. Flask is a lightweight and modular web application framework for Python. It is known for its simplicity, making it a popular choice for developing small to medium-sized web applications. Flask provides the necessary tools for routing, templating, and handling HTTP requests.

3.1 Methodology

- **Data Collection**

  Collect a diverse and representative dataset of retinal images from individuals with varying cardiovascular health statuses. The dataset should cover different demographics, age groups, and risk factors. Each image should be annotated to indicate whether the individual has a cardiovascular condition or is healthy, ensuring a balanced and comprehensive dataset.

- **Data Preprocessing**

  Preprocess the retinal images to ensure consistency and remove irrelevant information. This includes standardizing image resolution and color channels, normalizing pixel values, and enhancing image quality using techniques such as noise reduction, contrast adjustment, and image cropping. Data augmentation (rotation, flipping, zooming) may also be applied to increase dataset diversity and improve model robustness.

- **Model Development (CNN)**

  Design and implement a Convolutional Neural Network (CNN) architecture tailored for retinal image classification. Convolutional and pooling layers will be used to automatically extract important spatial features from the images, followed by fully connected layers for classification. The CNN model will be trained on the preprocessed dataset, learning to distinguish patterns and features associated with cardiovascular risks. Hyperparameters such as learning rate, batch size, and number of layers will be optimized to improve performance.

- **Model Evaluation**

  Evaluate the trained CNN model using standard performance metrics.

- **Testing and Deployment**

  Test the final CNN model on unseen retinal images to assess its real-world performance. Once validated, the model can be deployed within a Flask-based web application that allows users to upload retinal images and receive predictions regarding heart disease risk.

- **Dataset:**

  The dataset used in this project consists of retinal images categorized into different risk levels associated with diabetic retinopathy (DR) and other cardiovascular conditions. Diabetic retinopathy is a common complication of diabetes and is characterized by damage to the blood vessels in the retina. The dataset includes images representing various stages of DR, ranging from mild to severe, as well as images indicating the absence of DR. Additionally, the dataset may contain images representing proliferative diabetic retinopathy (PDR), the most advanced stage of DR characterized by the growth of abnormal blood vessels in the retina. Each retinal image in the dataset is annotated with its corresponding risk category, allowing for supervised learning algorithms to be trained for heart disease detection. This diverse dataset enables the development and evaluation of machine learning models aimed at leveraging retinal images for early detection and classification of heart diseases, providing valuable insights into the relationship between retinal characteristics and cardiovascular health.

3.2 Code snippets

3.2.1 Admin Login

```
@app.route('/login',methods=['POST','GET'])
def login():
    status=True
    if request.method=='POST':
        uname=request.form["email"]
        pwd=request.form["upass"]
        cur=mysql.connection.cursor()
        cur.execute("select * from admin where email=%s and password=%s",(uname,ct.md5(pwd)))
        data=cur.fetchone()
        if data:
            session['logged_in']=True
            session['username']=data["username"]
            flash('Login Successfully','success')
            return redirect('home')
        else:
            flash('Invalid Login credentials. Try Again','danger')
    return render_template("login.html",url = url)
```

3.2.2Registration

```
@app.route('/reg',methods=['POST','GET'])
def reg():
    status=False
    if request.method=='POST':
        name=request.form["uname"]
        email=request.form["email"]
        pwd=request.form["upass"]
        cur=mysql.connection.cursor()
        cur.execute("select * from user where username=%s",[name])
        data=cur.fetchone()
        if not data:
            cur.execute("insert into user(username,password,email) values(%s,%s,%s)",(name,pwd,email))
            mysql.connection.commit()
            cur.close()
            flash('Registration Successfully. Login Now...','success')
        else:
            flash('Username Exists...Kindly use different username','danger')
        return redirect('login')
    return render_template("login.html",status=status,url = url,data = session['username'])
```

3.2.3 Prediction

```
@app.route('/heart_predict', methods=['GET', 'POST'])
@is_logged_in
def heart_predict():
    global output_file
    if request.method == 'POST':
        f = request.files['file']
        file_path = os.path.join('static',secure_filename(f.filename))
        f.save(file_path)
        output_file = file_path
    return render_template('heart_predict.html',url=url, data = session['username'],filename = output_file)
```

CHAPTER 6

SOFTWARE TESTING

Testing is a critical phase in the software development life cycle that ensures the developed system meets its functional and non-functional requirements. For this project, testing verifies that the CNN-based heart disease prediction model, along with the supporting web application, performs reliably under different scenarios. It involves validating the correctness of data preprocessing, the accuracy of predictions, and the proper functioning of the system's interfaces. By conducting systematic testing, errors can be identified and corrected early, ensuring that the system is both robust and user-friendly for practical healthcare applications.

6.1 Unit Testing

Unit testing focuses on testing individual components or modules of the system in isolation to confirm that each part performs as expected. In this project, unit testing includes verifying the preprocessing functions, CNN model training module, image input handling, and prediction output functions. For example, preprocessing is tested to ensure images are correctly normalized, while the CNN layers are tested to verify that feature extraction is functioning properly. This helps in identifying and fixing bugs at the earliest stage of development, improving the overall reliability of the system.

6.2 Integration Testing

Integration testing ensures that the different modules of the project work together seamlessly. In this project, it checks the integration between the CNN model, the Flask-based web interface, and the MySQL database. For instance, the transition from image upload to preprocessing, model prediction, and result display is tested to ensure smooth data flow. Any mismatches in data handling or communication between modules are identified and resolved during this phase. Integration testing validates that the system behaves correctly as a combined unit rather than as isolated components.

6.2 System Testing

System testing evaluates the entire application as a whole to verify that it meets the specified requirements. For this project, system testing involves checking whether the application can handle multiple image uploads, generate accurate predictions, and provide outputs within acceptable response times. It also ensures that both Admin functions (such as dataset upload and model training) and User functions (such as registration, login, and prediction) operate correctly. System testing simulates real-world usage scenarios to confirm that the CNN-based heart disease prediction system is reliable, scalable, and ready for deployment in a healthcare environment.

6.3 Regression Testing

Regression testing plays a critical role in ensuring that updates or modifications to the project do not negatively impact existing functionalities. In this project, regression testing is performed whenever improvements are made to preprocessing functions, CNN architecture, or the integration between the web interface and the database. For instance, after retraining the CNN with new datasets or fine-tuning hyperparameters, regression testing verifies that previous functionalities, such as user registration, login, dataset upload, and prediction generation, continue to work correctly. This testing strategy ensures system stability, maintains prediction accuracy, and safeguards user experience throughout iterative development.

6.4 Test Cases

Table 6.1 Test cases

TC 01	Data Collection	To import Data from location	Import from file location	Valid location and file name	Import data successful	Import data successful	Pass
	Data Collection	To import Data from location	Import from file location	Invalid location and	Import data unsuccessful	Import data unsuccessful	Pass

				file name			
TC 02	Data pre-processing	To verify that data has pre-processed	Enter the data file imported	Valid detail	pre-processing successfully	pre-processing successfully	Pass
	Data pre-processing	To verify that data has pre-processed	If invalid data file imported	Invalid detail	Pre-processing unsuccessfuly	Pre-processing unsuccessfuly	Pass
TC 03	Train the data	To train data	Implementalgorithms	Successoutput	Training successful	Training successful	Pass
TC 04	Detection	Test the trained algorithm	Implement best algorithm to dataset	Success output	Prediction successful	Prediction successful	Pass

CHAPTER 7

RESULTS

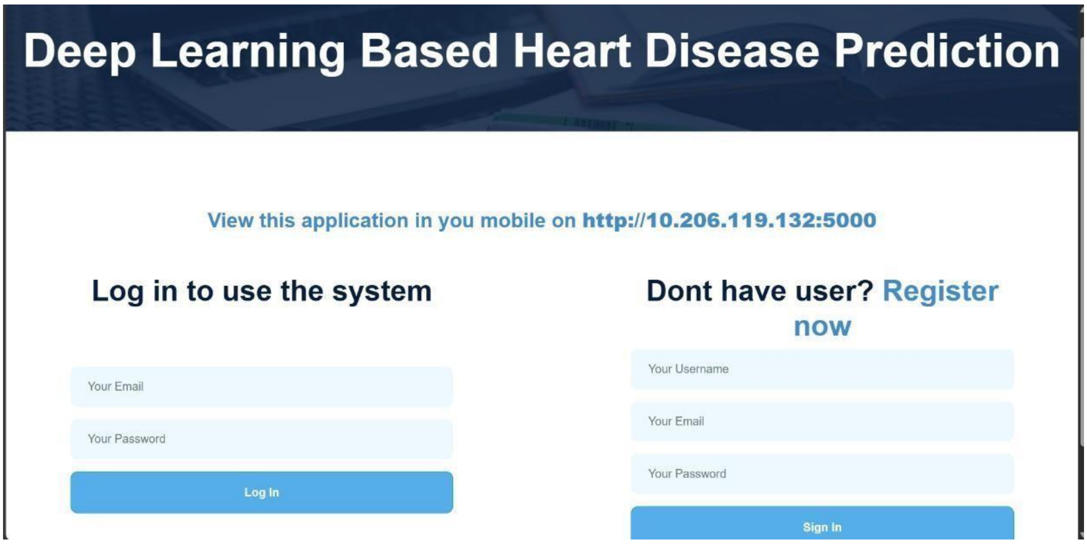

Fig 7.1 User login page

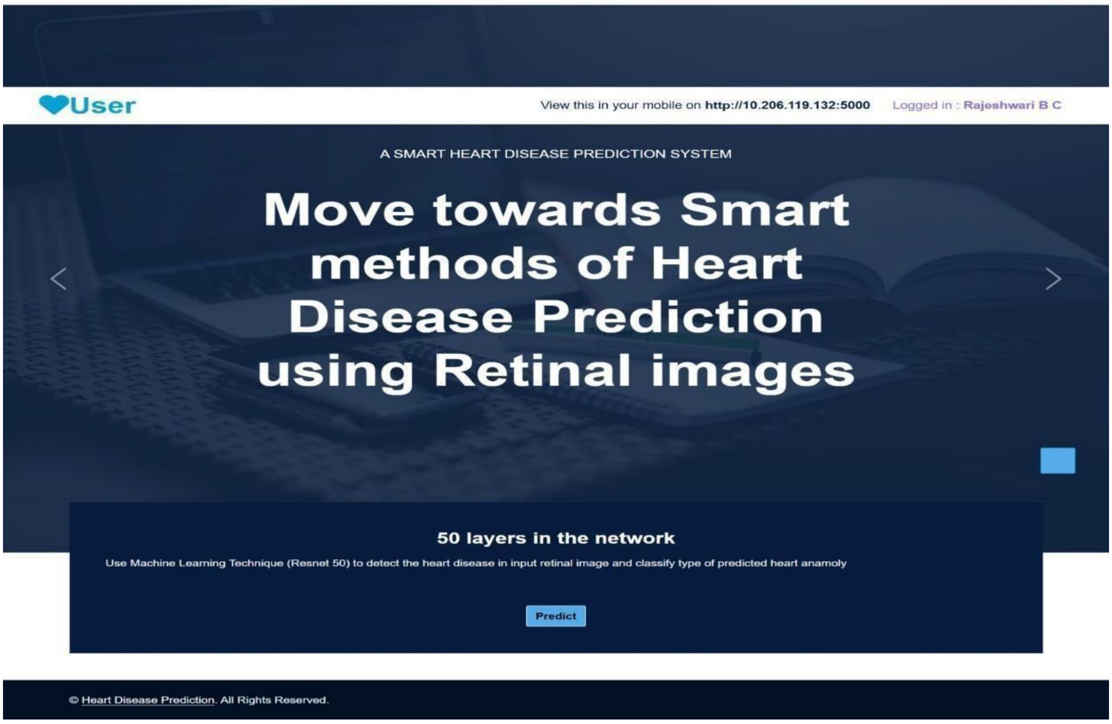

Fig 7.2 Home page

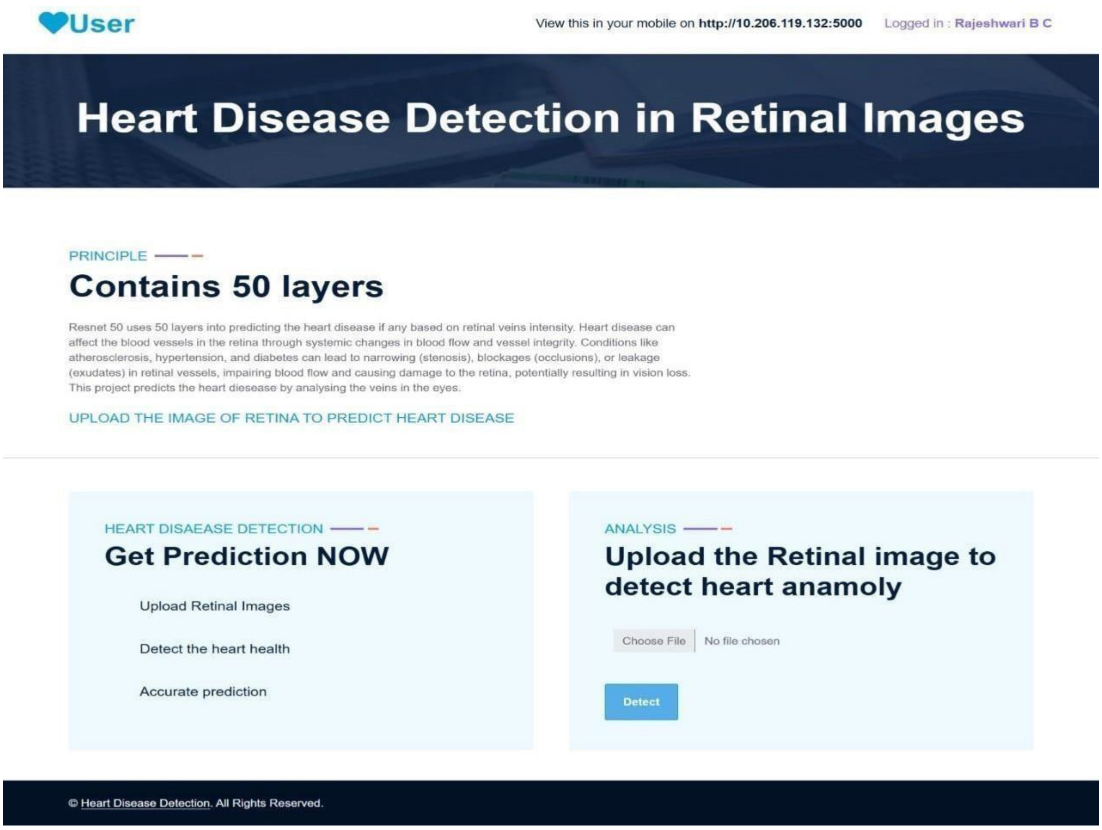

Fig 7.3 Uploading a retinal image

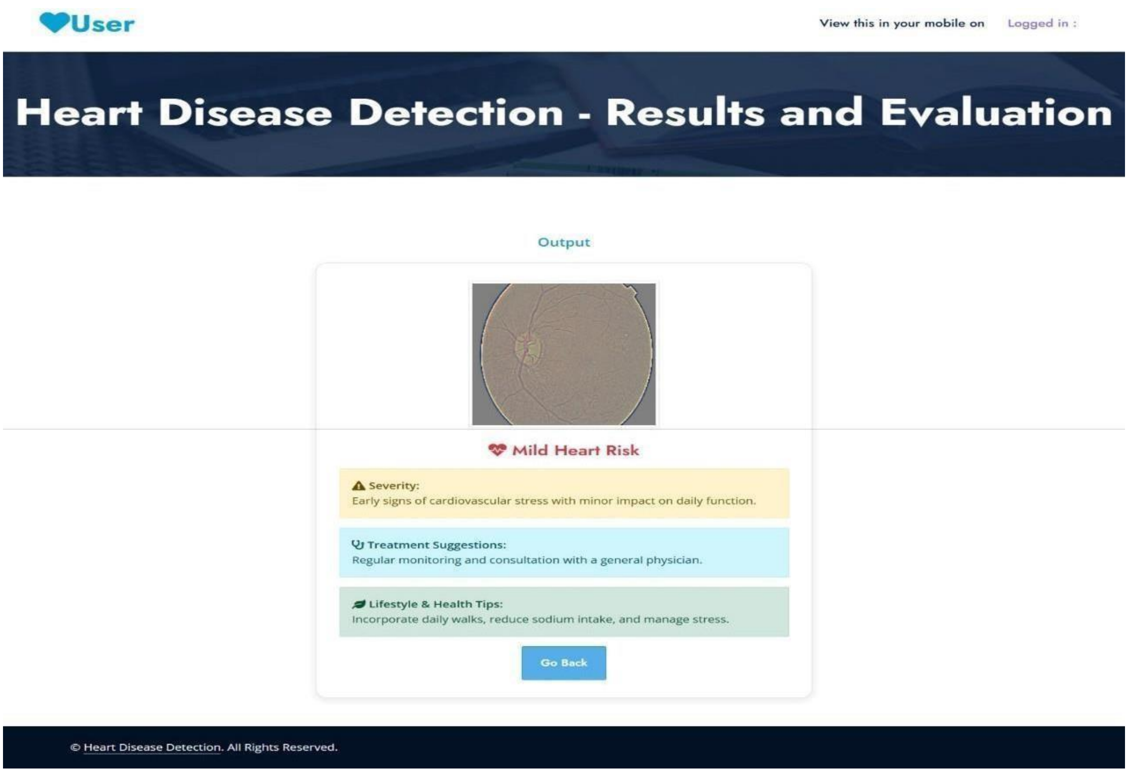

Fig 7.4 Predicting heart disease risk

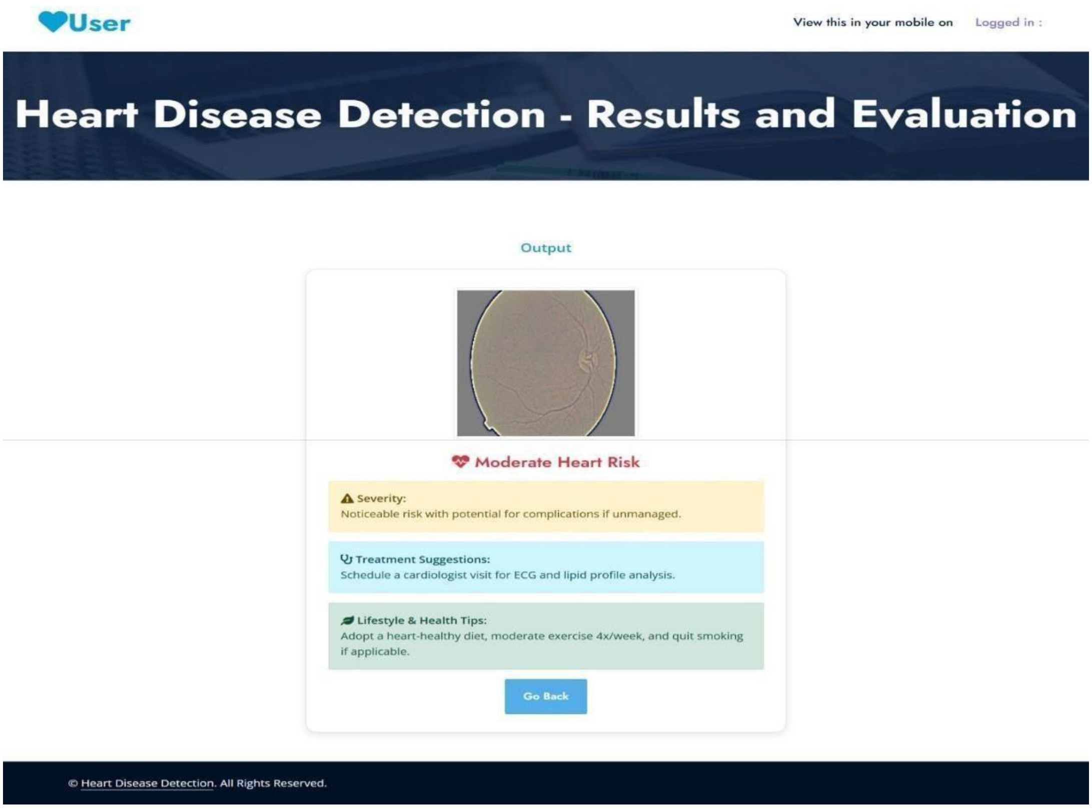

Fig 7.5 Predicting heart disease risk

CONCLUSION

This project demonstrates the potential of Convolutional Neural Networks (CNNs) in analyzing retinal images for the early detection of heart disease. By leveraging the spatial feature extraction capabilities of CNNs, the system is able to identify subtle patterns and abnormalities in retinal blood vessels that are often linked to cardiovascular conditions. The non-invasive nature of retinal imaging, combined with the efficiency of deep learning, provides a promising alternative to conventional diagnostic methods that are often invasive, costly, or time-consuming.The results highlight that CNN-based models can achieve high accuracy, sensitivity, and specificity in detecting heart disease risks from retinal images. This approach not only aids in early diagnosis but also has the potential to be integrated into telemedicine platforms and routine clinical screenings, making cardiovascular health assessment more accessible.Overall, the project contributes to the growing field of AI-driven healthcare by providing a cost-effective, reliable, and scalable solution for heart disease risk prediction. With further refinement, larger datasets, and clinical validation, such a system could play a critical role in supporting healthcare professionals, enabling proactive interventions, and ultimately reducing the global burden of cardiovascular diseases.

FUTURE ENHANCEMENT

Integrate additional modalities of data, such as patient demographics, clinical history, genetic information, or other medical imaging modalities like echocardiography or MRI scans. A key future enhancement of this project is to develop a multi-modal cardiovascular risk prediction system by combining retinal images with clinical data such as age, blood pressure, cholesterol levels, ECG reports, and lifestyle factors. Integrating advanced deep learning models like Vision Transformers or hybrid CNN-Transformer architectures can further improve feature extraction and prediction accuracy. The system can also be enhanced by incorporating real-time screening through mobile or fundus camera–based applications, enabling early detection in remote or rural areas. Additionally, applying explainable AI (XAI) techniques such as Grad-CAM can help clinicians understand which retinal regions influence predictions, increasing trust and clinical adoption.

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